

BEC503 Digital Communication — Assignment 1 — Signal space representation and optimum reception

Module 1: Bandpass Signals to Equivalent Lowpass and Signalling over AWGN Channels

Field	Value
Maximum marks	20
Weightage	10% of CIE
Issue date	< <TBD> >
Due date	< <TBD> >
Items	8 problems + 3 theory questions
CO / Bloom coverage	CO1: 8, CO2: 3 · L1–L2 0, L3–L4 9, L5–L6 2

Objective

Move fluently between the three descriptions of a bandpass signal — waveform, complex envelope, and point in signal space — and derive the optimum receiver from the geometry rather than quoting it.

Learning outcomes

- Student will be able to construct the pre-envelope and complex envelope of a given bandpass signal and recover the original from either.
- Student will be able to apply Gram-Schmidt to an arbitrary signal set and state the dimension of the space it spans.
- Student will be able to compute average energy, minimum distance and decision regions for a given constellation.
- Student will be able to justify the correlation and matched-filter structures as implementations of the same projection.

Part A — Problems (17 marks)

P1. For $s(t) = 4 \cos(2 \pi f_c t) - 3 \sin(2 \pi f_c t)$, find the in-phase and quadrature components, the complex envelope in both rectangular and polar form, the natural envelope and the instantaneous phase. State the peak-to-average power ratio. [C01 | L3 | 3 marks]

P2. Prove that a signal and its Hilbert transform are orthogonal over all time. Then show by direct integration that $\cos(2 \pi f_0 t)$ and $\sin(2 \pi f_0 t)$ are orthogonal over an integer number of periods but not over an arbitrary interval, and quantify the residue for an interval of 1.25 periods. [C01 | L4 | 3 marks]

P3. Apply the Gram-Schmidt procedure to the four signals $s_1 = (1,1,0,0)$, $s_2 = (1,0,1,0)$, $s_3 = (0,1,1,0)$ and $s_4 = (2,1,1,0)$, treating each as a sequence of four unit-duration pulse amplitudes. Report the orthonormal basis, the dimension N , and identify which signal contributed no new dimension and why. [C01 | L4 | 3 marks]

- P4.** For an 8-PSK constellation of average energy E , compute the minimum Euclidean distance and compare it with QPSK at the same average energy. Express both as a multiple of $\sqrt{E}$ and state the resulting difference in required SNR in decibels. [C02 | L4 | 2 marks]
- P5.** Sketch the decision regions for a 4-point constellation at (0,0), (2,0), (0,2) and (2,2) with equal priors. Then redraw them for priors 0.4, 0.2, 0.2, 0.2 and state, with the relevant term of the MAP rule, how the boundaries move. [C01 | L4 | 2 marks]
- P6.** A rectangular pulse of amplitude A and duration T is received in AWGN of two-sided density $N_0/2$. Derive the matched filter impulse response, sketch its output, and compute the peak SNR. Then compute the SNR loss in decibels if the output is sampled at $0.8T$ instead of T . [C01 | L4 | 1 marks]
- P7.** Simulate or hand-compute 20 observations of a BPSK system at $E_b/N_0 = 4$ dB using any tool you like. Report the number of errors, compare against the theoretical $Q(\sqrt{2 E_b/N_0})$, and comment on why 20 trials is far too few to confirm the theory. [C02 | L5 | 2 marks]
- P8.** Explain, with a diagram, what happens to the observation vector of a correlation receiver when the integrator is not reset between symbols. Predict the appearance of the resulting constellation after 10 symbols. [C01 | L5 | 1 marks]

Part B — Theory (3 marks)

- T1.** Argue whether the theorem of irrelevance would still hold if the noise were Gaussian but coloured. Identify precisely which step of the argument fails and what pre-processing restores it. [C01 | L4 | 1 marks]
- T2.** The matched filter maximises peak SNR but does not minimise intersymbol interference. Explain the tension between these two objectives and how raised-cosine pulse shaping resolves it. [C02 | L4 | 1 marks]
- T3.** The complex envelope is described as an exact representation rather than an approximation. State the one condition on f_c and W that makes this true, and describe what goes wrong when it is violated. [C01 | L4 | 1 marks]

Submission guidelines

- Format: PDF, single file
- File name: <USN>_BEC503_Assignment1.pdf
- Channel: LMS upload
- Late policy: 10% per day, nothing after 5 days
- Every numerical answer must carry units and a stated reference direction.

Assessment rubric

Criterion	Weight	Excellent	Good	Satisfactory	Needs improvement
Correctness of derivations	8	All derivations correct with every step justified and assumptions stated.	Derivations correct with minor gaps in justification.	Main results reached but several steps unjustified or algebraically slipped.	Results quoted without derivation or substantially incorrect.

Criterion	Weight	Excellent	Good	Satisfactory	Needs improvement
Numerical accuracy and units	5	All numbers correct, units consistent, dB conversions right.	One or two arithmetic slips that do not change conclusions.	Several numerical errors or inconsistent units.	Numbers largely wrong or units absent.
Diagrams and signal space sketches	4	Constellations, decision regions and filter outputs drawn accurately and labelled.	Diagrams correct but sparsely labelled.	Diagrams present but inaccurate in scale or boundaries.	Diagrams missing or misleading.
Interpretation and engineering judgement	3	Explains what each result means for a real receiver and where the assumptions break.	Some interpretation offered beyond the mathematics.	Minimal interpretation; results left unexplained.	No interpretation attempted.
Total	20				

Course outcome attainment

Course outcome	Marks carried
CO1	15
CO2	5
Total	20

Academic integrity

Individual submission. Identical submissions score zero for both parties.